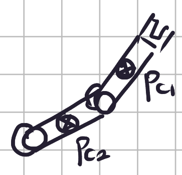
Exercise 1

Consider a 2R planar robot, with the two links of length l_1 and l_2, having the actuating motors mounted on the axes of the two revolute joints. Each motor delivers its torque to the driven link through an elastic transmission, modeled as a torsional spring of stiffness $k_i > 0$, for $i = 1, 2$. The robot has no motion reduction elements. In the ith motor-link assembly, for $i = 1, 2$, let m_{m_i} and I_{m_i} be, respectively, the balanced mass and the inertia of the rotor of the motor around its spinning axis, and m_i, d_{c_i}, and I_i the link mass, the distance of the center of mass of the link from the preceding joint axis, and the link inertia around its center of mass. Using as generalized coordinates the angle θ_{m_i} of the rotor of motor i w.r.t. the preceding link axis, and the angle θ_i of link i w.r.t. the preceding link axis, for $i = 1, 2$, define the 4×4 inertia matrix $\boldsymbol{M}(\boldsymbol{q})$ of the robot, where $\boldsymbol{q} = (\boldsymbol{\theta}^T\ \boldsymbol{\theta}_m^T)^T$. State explicitly any simplifying assumption that you may wish to use. Moreover, find a linear parametrization of the inertial term $\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} = \boldsymbol{Y}_M(\boldsymbol{q}, \ddot{\boldsymbol{q}})\,\boldsymbol{a}$ of the robot dynamic model in terms of a minimal set of p suitable dynamic coefficients a_i, $i = 1, \dots, p$.



$$p_{c_1} = \begin{cases} l_1/2\, c_1 \\ l_1/2\, s_1 \end{cases} \qquad p_{c_2} = \begin{cases} l_1 c_1 + \frac{l_2}{2} c_{12} \\ l_1 s_1 + \frac{l_2}{2} s_{12} \end{cases}$$

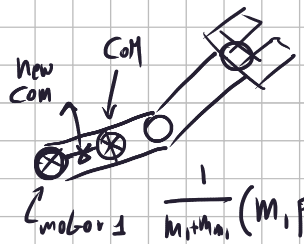
no red. ratio

For the kinetick energy of the motors: $T_m = \frac{1}{2}\left(I_{m_1} \overset{1}{n_{r_1}} \dot\theta_{m_1}^2 + I_{m_2} \overset{1}{n_{r_2}} \dot\theta_{m_2}^2\right)$

$$\Rightarrow T_m = \frac{1}{2} I_{m_1}\dot\theta_{m_1}^2 + \frac{1}{2} I_{m_2}\dot\theta_{m_2}^2 = \frac{1}{2}\dot\theta_m M_m \dot\theta_m \Rightarrow M = \begin{pmatrix} ? & 0 & 0 \\ 0 & 0 & \\ 0 & 0 & M_m \end{pmatrix} = \begin{pmatrix} // & // & 0 & 0 \\ // & // & 0 & 0 \\ 0 & 0 & I_{m_1} & 0 \\ 0 & 0 & 0 & I_{m_2} \end{pmatrix}$$

$e_4\ e_5$

We move the motors weigths to the links.



the CoM of link 1 is moved near the base due to the actuator. The new CoM is

$$\frac{1}{m_1+m_{m_1}}\left(m_1 p_{c_1} + m_{m_1} p_{m_1}\right) \quad \text{but } p_{m_1} = 0 \text{ (base) so}$$

$$\Rightarrow p'_{c_1} = \frac{m_1}{m_1+m_{m_1}} p_{c_1} \Rightarrow v_{c_1} = \frac{m_1 l_1}{2(m_1+m_{m_1})}\begin{cases} -s_1\dot\theta_1 \\ c_1\dot\theta_1 \end{cases} \Rightarrow T_1 = \frac{1}{2}\left(\frac{l_1 m_1(m_1+m_{m_1})}{2(m_1+m_{m_1})} + I_1\right)\dot\theta_1^2$$

$$p_{m_2} = \begin{cases} l_1 c_1 \\ l_1 s_1 \end{cases} \quad \text{so} \quad \overset{\text{new}}{CoM_2} = \frac{1}{m_2+m_{m_2}}\left(m_2 p_{c_2} + m_{m_2} p_{m_2}\right) =$$

$$\frac{1}{m_2+m_{m_2}}\begin{pmatrix} m_2\left(l_1 c_1 + \frac{l_2}{2} c_{12}\right) + m_{m_2} l_1 c_1 \\ m_2\left(l_1 s_1 + \frac{l_2}{2} s_{12}\right) + m_{m_2} l_1 s_1 \end{pmatrix} \Rightarrow$$

$\rightarrow \dot\theta_i = \dot q_i$

$$v_{c_2} = \frac{1}{m_2+m_{m_2}}\begin{pmatrix} -l_1 s_1(m_2+m_{m_2})\dot q_1 - \frac{l_2}{2} m_2 s_{12}(\dot q_1+\dot q_2) \\ l_1 c_1(m_2+m_{m_2})\dot q_1 + \frac{l_2}{2} m_2 c_{12}(\dot q_1+\dot q_2) \end{pmatrix}$$

$$T_2 = \frac{1}{2} I_2(\dot\theta_1+\dot\theta_2)^2 + \frac{1}{2}\Big(l_1^2(m_2+m_{m_2})^2\dot q_1^2 + \frac{l_2^2}{4} m_2^2(\dot q_1^2+\dot q_2^2+2\dot q_1\dot q_2)$$
$$+ 2 l_1 \frac{l_2}{2} m_2(m_2+m_{m_2})(\dot q_1^2+\dot q_1\dot q_2)\cos(\theta_2-\theta_1)\Big)$$

$$M_{\theta 11} = \overbrace{I_2}^{} + \overbrace{\ell_1^2 (m_2 + m_{m2})^2}^{e_3} + \overbrace{\frac{\ell_2^2}{4} m_2^2}^{e_1} + \overbrace{\ell_1 \ell_2 m_2 (m_2 + m_{m2})}^{e_2} \cos(\theta_2 - \theta_1) + \frac{\ell_1 m_1 (m_1 + m_{m1})}{2(m_1 + m_{m1})} + I_1 =$$

$$M_{\theta 12} = \overbrace{\frac{\ell_2^2}{4} m_2^2}^{} + \overbrace{\ell_1 \frac{\ell_2}{2} m_2 (m_2 + m_{m2})}^{} \cos(\theta_2 - \theta_1) + \overbrace{I_2}^{} = e_1 + \frac{1}{2} e_2 \cos(\theta_2 - \theta_1)$$

$$M_{\theta 22} = I_2 + \frac{\ell_2^2}{4} m_2^2 = e_1 \qquad M_{\theta 11} = e_1 + e_3 + e_2 \cos(\theta_2 - \theta_1)$$

$$\Rightarrow M = \begin{pmatrix} e_1 + e_3 + e_2 \cos(\theta_2 - \theta_1)\ \ddot\theta_1 & e_1 + \frac{1}{2} e_2 \cos(\theta_2 - \theta_1)\ \ddot\theta_2 & 0 & 0 \\ e_1 + \frac{1}{2} e_2 \cos(\theta_2 - \theta_1)\ \ddot\theta_1 & e_1\ \ddot\theta_2 & 0 & 0 \\ 0 & 0 & e_4 \ddot\theta_{m1} & 0 \\ 0 & 0 & 0 & e_5\ \ddot\theta_{m2} \end{pmatrix}$$

$$\boxed{\begin{matrix} \ddot\theta_1 + \ddot\theta_2 & \ddot\theta_1 \cos(\theta_2 - \theta_1) \frac{3}{2} & \ddot\theta_1 & 0 & 0 \\ \ddot\theta_1 + \ddot\theta_2 & \ddot\theta_1 \cos(\theta_2 - \theta_1) \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \ddot\theta_{m1} & 0 \\ 0 & 0 & 0 & 0 & \ddot\theta_{m2} \end{matrix}} = Y$$

Exercise 2

Consider a 4R planar robot with all links of equal length $\ell = 0.2$ [m]. The robot is in the DH configuration $\boldsymbol{q} = (\ 0\ \ \pi/2\ \ 0\ \ \pi/2\ )^T$ [rad] and at rest ($\dot{\boldsymbol{q}} = \boldsymbol{0}$). In this state, we should assign to the end-effector a desired linear acceleration $\boldsymbol{a} = (\ 5\ \ 0\ )^T$ [m/s^2]. The joint accelerations are taken as input commands, and are bounded as $|\ddot q_i| \le A_i$, $i = 1, \ldots, 4$, with the limits $A_1 = 9$, $A_2 = 6$, $A_3 = 4$, and $A_4 = 2$ [rad/s^2]. Find, if possible, a feasible joint acceleration $\ddot{\boldsymbol{q}} \in \mathbb{R}^4$ that executes instantaneously the desired Cartesian task, while satisfying these hard bounds. A solution with a lower norm is preferred, and could be obtained by a straightforward variation of the SNS method moved to the acceleration level.

$S_1 = 0 \qquad S_{12} = 1 \qquad S_{123} = 1 \qquad S_{1234} = 0$

$C_1 = 1 \qquad C_{12} = 0 \qquad C_{123} = 0 \qquad -1$

$* \ \ddot p = J \ddot q + \underbrace{\dot J \dot q}_{=0} \rightarrow \ddot p = J \ddot q$

The Jacobian is: $J = \frac{2}{10} \begin{pmatrix} 2 & 2 & 1 & 0 \\ 0 & -1 & -1 & -1 \end{pmatrix}$

the pseudo inverse would provide $\ddot q = J^{\#} \begin{pmatrix} 5 \\ 0 \end{pmatrix} = (8.333 \quad 4.166 \quad 0 \quad -4.166)^T$

violates the bound on $\ddot q_4$. The SNS procedure is

$$\ddot q = s (JW)^{\#} a + (I - (JW)^{\#} J) \ddot q_N \qquad \ddot q_N = (9 \quad 6 \quad 4 \quad -2)$$

$$W = \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 0 \end{pmatrix} \leftarrow \text{saturate } \ddot q_4 \qquad \text{let } s = 0.871$$

$$\Rightarrow \ddot q_{SNS} = (8.9 \quad 1.975 \quad 0.025 \quad -2)^T \quad \text{the task error is:}$$

$$\| a - J \ddot q_{SNS} \| = 0.645$$

Exercise 3

In a visual servoing scheme, n point features with coordinates (u_i, v_i), for $i = 1, \ldots, n$, can be extracted from the image. Define the 2×6 interaction matrix $\bar{\boldsymbol{J}}$ between the 6D vector of linear velocity $\boldsymbol{V} \in \mathbb{R}^3$ and angular velocity $\boldsymbol{\Omega} \in \mathbb{R}^3$ of the camera and the time derivative of the coordinates $(\bar{u}, \bar{v})$ of the average position of the n point features in the image plane. State all variables that matrix $\bar{\boldsymbol{J}}$ depends upon.

short notation: $\sum_{i=1}^{n} := \sum$

The interaction matrix is:

$$J_p = \begin{pmatrix} -\left(\sum z_i\right)^{-1} \lambda n & 0 & \sum \frac{u_i}{z_i} & \frac{\sum u_i v_i}{\lambda n^2} & -\left(\lambda + \frac{\sum u_i}{\lambda n^2}\right) & \frac{1}{n} \sum v_i \\ 0 & -\left(\sum z_i\right)^{-1} \lambda n & \sum \frac{v_i}{z_i} & \lambda + \frac{\sum v_i}{\lambda n^2} & -\frac{\sum u_i v_i}{\lambda n^2} & -\frac{1}{n} \sum u_i \end{pmatrix}$$

J_p depends on all the points u_i, v_i $i = 1 \cdots n$, the focal length λ and the depths z_i (to be estimated) and the number of points n.

Exercise 4

For a robot with n degrees of freedom, partition the generalized coordinates as $\boldsymbol{q} = (\boldsymbol{q}_a, \boldsymbol{q}_b)$, where $\boldsymbol{q}_a$ has n_a components, $\boldsymbol{q}_b$ has n_b components, and $n_a + n_b = n$. Provide the explicit expressions of the n_a-dimensional reduced robot dynamics and of the constraint-preserving forces $\boldsymbol{\lambda} \in \mathbb{R}^{n_b}$, when the geometric constraint $\boldsymbol{h}(\boldsymbol{q}) = \boldsymbol{q}_b - \boldsymbol{q}_{b,d} = \boldsymbol{0}$ is imposed at all times, with $\boldsymbol{q}_{b,d}$ being constant.

$$A = \frac{\partial h}{\partial q} = \begin{array}{c} \\ n_b \end{array} \begin{array}{cc} n_a & n_b \\ \left(0 \right. & \left. I \right) \end{array} \Rightarrow D = \begin{array}{c} \\ n_a \end{array} \begin{array}{cc} n_a & n_b \\ \left(I \right. & \left. 0 \right) \end{array} \quad \begin{pmatrix} A \\ D \end{pmatrix} = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}$$

$$\Rightarrow D\dot{q} = \left(\dot{q}_1 \; \dot{q}_2 \cdots \dot{q}_{na}\right)^T \qquad \begin{pmatrix} A \\ D \end{pmatrix}^{-1} = \begin{pmatrix} & I \\ I & \end{pmatrix} \Rightarrow E = \begin{pmatrix} 0 \\ I \end{pmatrix} \quad G = \begin{pmatrix} I \\ 0 \end{pmatrix}$$

λ is given by $\boldsymbol{\lambda} = E^T(MG\dot{\mathbf{u}} - M(E\dot{A} + G\dot{D})G\mathbf{u} + \mathbf{c} + \mathbf{g} - \boldsymbol{\tau})$.

$$\Rightarrow \lambda = E^T\left(MG\left(\dot{q}_1 \cdots \dot{q}_{na}\right) + c + g - \tau\right)$$